M.Tech · Fundamentals of Artificial Intelligence

Mini Project Report

AI-Based University Enquiry Chatbot

using Knowledge Representation and Natural Language Processing

Project topic No. 29 · Suggested technique: Knowledge Representation + NLP

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Live chatbot: https://manikandanayyakkannu.github.io/FAI-Mini-Project/

Submitted along with this report: the solution notebook (University_Enquiry_Chatbot.ipynb), the chatbot website (index.html and web/), the Python engine (chatbot/) and the data files (data/).

Abstract

In this project we built an enquiry chatbot for a fictional university, Greenfield University. Students can ask it about programmes, fees, eligibility, admissions, the hostel, scholarships and placements. The university data is stored as frames that inherit from each other (a small semantic network). Personal questions such as "Am I eligible?" or "What will my fee be after the scholarship?" are answered by forward-chaining production rules. In front of this sit an NLP pipeline, a dialogue manager for follow-up questions, and a vector store for FAQ topics that are not in the frames. We tested it on 105 labelled questions. Our chatbot answered 92.4% of them correctly (intent macro-F1 0.93). On the same knowledge, an NLP + vector-retrieval chatbot managed 38.1% and a keyword FAQ-matching chatbot 31.4%. On a separate held-out set, which we ran only once at the end, it scored 80.0%. Answers take a few milliseconds (mean 1.6 ms) and every answer can be explained. The finished chatbot is a website that runs completely inside the browser, so anyone can try it from a link.

Keywords: knowledge representation, frames, semantic network, forward chaining, production rules, natural language processing, chatbot, vector store

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1. Problem Definition

During admission season a university's office gets the same questions again and again, by phone, by e-mail and at the front desk: "What is the fee for B.Tech CSE?", "Am I eligible for M.Tech AI with 6.8 CGPA?", "When is the last date to apply?", "Is there a hostel for girls?". Answering them takes up a lot of staff time. Nobody answers after office hours, and two staff members do not always give the same answer.

Our aim was to build an enquiry chatbot that:

- understands free-form English questions, including paraphrases, abbreviations (B.E, M.Tech, CGPA, plus two) and spelling mistakes;
- answers factual questions from a structured knowledge base of the university;
- reasons about personal questions, such as eligibility for a programme or the fee after scholarships, and explains every decision;
- follows the conversation ("How long is it?" after a question about M.Tech AI) and asks for missing details;
- answers free-text FAQs (ragging, Wi-Fi, refunds) through semantic retrieval, and politely refuses out-of-scope questions.

Scope and assumptions: the knowledge base describes a fictional institution, "Greenfield University", with 9 programmes (B.Tech CSE, AI & DS, ECE and Mechanical; M.Tech AI and Data Science; MBA; MCA; Ph.D.), 6 departments, 5 facilities, the admission process and the scholarship policy. We made up all of this data for the project. The chatbot is text-based, works in English and runs in any web browser without a server.

2. AI Problem Formulation

The chatbot is modelled as a knowledge-based agent. It perceives a sentence, updates its internal state (the dialogue context) and acts by choosing the answer that its knowledge base and inference rules support.

2.1 PEAS description

PEAS element	Description
Performance measure	Answer accuracy, intent precision / recall / F1, correct refusal of out-of-scope questions, response time, explainability
Environment	Prospective students and parents typing questions; the university knowledge (frames, rules, FAQ passages)
Actuators	Text answers, clarification questions, explanation traces
Sensors	The user's typed sentence

2.2 Properties of the task environment

Property	Value	Reason
Observability	Partially observable	The user's intention and personal facts are hidden and must be inferred from text
Determinism	Stochastic	The same need can be phrased in many different ways
Episodes	Sequential	Follow-up questions depend on earlier turns
Dynamics	Static	The knowledge does not change during a conversation

State space	Discrete	Finite sets of intents, entities and facts
Agents	Single agent	One chatbot serves one user per session

2.3 Formal formulation

- **Knowledge base (KB):** a set of frames F = {University, Program, UGProgram, PGProgram, btech_cse, ..., hostel, ...} linked by is_a edges. Slot lookup uses inheritance: value(f, s) = f.s if the frame stores slot s, otherwise value(parent(f), s).
- **Percept to query mapping:** the NLP pipeline maps a sentence x to a structured query q = (intent, entities), where intent belongs to 18 intents (program_fee, eligibility, hostel, ...) and the entities are programme, qualification, percentage, CGPA, entrance exam, category, subjects and so on.
- **Working memory (WM):** for a personal question, WM = facts about the programme (from the frames) plus facts about the applicant (from the sentence).
- **Operators (rules):** production rules of the form IF c1 AND c2 AND ... THEN add/modify fact (16 eligibility rules and 10 scholarship rules).
- **Inference and goal test:** forward chaining fires every applicable rule once (refraction) until a goal fact (academic_eligible, net_fee) is derived or no rule can fire.
- **Fallback:** if no intent is confident, the query is embedded as a vector and the closest FAQ passage is returned when its cosine similarity is above a threshold; otherwise the agent replies "out of scope".

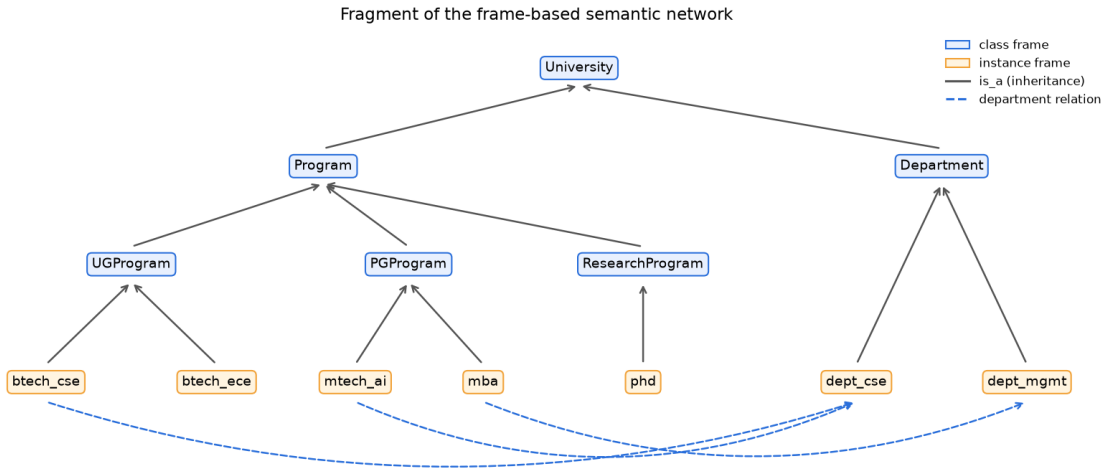


Figure 1. Fragment of the frame-based semantic network (is_a inheritance and department relations).

3. Selection and Explanation of the AI Technique

The suggested technique for this topic was Knowledge Representation + NLP. We ended up with five parts, each from classical AI. The table says what each part does and why we picked it:

Component	Technique used	Why it was chosen
Knowledge representation	Frames with is_a inheritance, which also form a semantic network of (subject, predicate, object) triples	University data is naturally hierarchical (Programme > UG/PG > B.Tech CSE). Inheritance stores duration, documents and deadlines once per class, so the KB is small and consistent.
Reasoning	Production rules + forward chaining with refraction and an explanation trace	Eligibility and scholarship policies are IF-THEN rules. Forward chaining is data-driven: it starts from the applicant's facts and derives the conclusion. Each fired rule becomes a sentence of explanation.

NLP	Normalisation, tokenisation, spelling correction (edit-distance similarity), synonym mapping, stop-word removal, stemming, regex entity extraction, hybrid intent classifier	Easy to follow and explain; needs only 109 example sentences; no GPU or downloaded model; handles Indian academic abbreviations (B.E, M.Tech, CGPA, plus two).
Retrieval fallback	Vector store: TF-IDF word + character n-gram embeddings, cosine similarity, top-k search	Free-text FAQ passages (ragging, Wi-Fi, refunds) are not worth modelling as frames. The embedder is pluggable, so a neural sentence embedding can be used without code changes.
Dialogue management	Context memory + slot filling	Supports follow-up questions and asks for missing facts (marks, qualification) before running the rules.

3.1 Hybrid intent classifier

For each intent i the classifier computes score(i) = 0.6 x max cosine similarity between the query and the intent's example sentences (TF-IDF vectors of the pre-processed text) + 0.4 x keyword evidence (1 if a high-precision trigger word of the intent is present). The best intent wins. If even the best score is below 0.30, the query is passed to the vector store or answered as out of scope. A domain-relevance gate rejects questions that contain no word known to the university domain, such as "tell me a joke".

3.2 Forward chaining with explanations

For the question "I did B.E ECE with 6.8 CGPA and qualified GATE - am I eligible for M.Tech AI?" the working memory starts with the programme facts from the frame (minimum 60%, accepted qualifications, entrance exams) and the applicant facts from NLP (qualification = btech_ece, cgpa = 6.8, exam = gate). The engine then fires:

```
[R1]  Converted CGPA 6.8 to 64.6% using the standard formula (CGPA x 9.5).
[R2]  The minimum required aggregate for M.Tech Artificial Intelligence is 60%.
[R4]  Your qualification (B.Tech ECE) is accepted for M.Tech Artificial Intelligence.
[R6]  Your score 64.6% meets the cut-off of 60%.
[R11] There is no specific subject requirement for M.Tech Artificial Intelligence.
[R12] Conclusion: you satisfy the academic eligibility for M.Tech Artificial Intelligence.
[R14] Your GATE score is accepted for admission.
[R16] Because you qualified GATE you are eligible for a stipend: Rs. 12,400 per month.
```

3.3 Why not a pure machine-learning or LLM chatbot?

We did consider using a large language model, but decided against it. For admissions the numbers (fees, cut-offs) have to be exactly right, the same every time, and explainable. In our system every answer can be traced back to a frame slot or a rule. A language model can make up a fee or a cut-off, it cannot properly justify an eligibility decision, and it needs a paid API or a GPU. Our approach has its own cost: someone has to write the knowledge by hand. We discuss this in Section 8.

4. Python Implementation

We first wrote the chatbot as a small Python 3.12 package. The solution notebook uses it for all the experiments, and it was used to produce every number in this report. It needs numpy, pandas, scikit-learn (TF-IDF and metrics) and matplotlib. The NLP and reasoning parts use only the Python standard library.

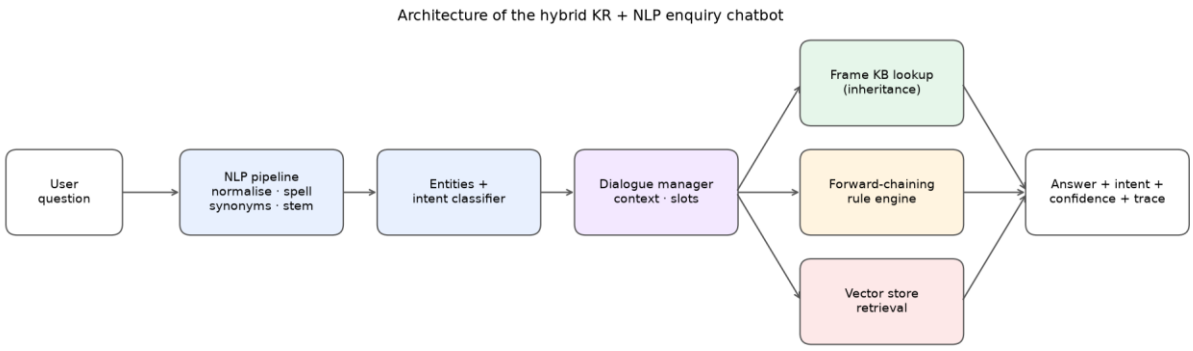


Figure 2. Architecture of the hybrid KR + NLP enquiry chatbot.

Module	Responsibility
knowledge_base.py	Frames, is_a inheritance (get, get_with_source, resolved), semantic-network triples and pattern queries
nlp.py	Normalisation of abbreviations, tokenisation, spelling correction, synonyms, stop words, stemming, entity extraction
intent_classifier.py	Hybrid TF-IDF similarity + keyword intent classifier with a confidence threshold
reasoning.py	Generic forward-chaining production-rule engine (conditions, actions, refraction, explanations)
vector_store.py	In-memory vector store with a pluggable embedder (TF-IDF default; sentence-transformers optional)
answers.py / faq_builder.py	Natural-language answer templates filled from frames; FAQ corpus for retrieval and the baselines
engine.py	UniversityChatbot: dialogue manager (context, slot filling), routing to KB / rules / vector store, trace
baselines.py / evaluation.py	Alternative approaches for comparison; test harness and metrics
data/*.json, data/*.csv	Knowledge base (29 frames), rules (26), intents (18), free-text FAQs (15), test sets

4.1 The chatbot website

For the demonstration we wanted a link that anyone could open, with nothing to install and no terminal. So we rewrote the same engine in plain JavaScript, one file for each Python module. It runs entirely inside the browser and is hosted for free on GitHub Pages. There is no server and no API key, and both versions read the same data/*.json files. To make sure the website behaves exactly like the Python version, we put all 154 test conversations through both. The answers, intents, answer sources and confidence scores came out identical, so the results in Sections 6 and 7 also hold for the website.

File	What it contains
index.html	The chat page: sample questions on the left, the conversation in the middle, the "How I answered" panel on the right
web/css/style.css	Layout and colours (light and dark theme, works on phones)
web/js/knowledge-base.js, nlp.js, intent-classifier.js, reasoning.js, answers.js, similarity.js	The engine, one file per Python module
web/js/chatbot.js	Dialogue manager: decides whether the KB, the rules or the vector store answers
web/js/app.js	Everything on the screen: messages, suggestions, explanation panel, guided demos
tests/engine-test.html	Runs both test sets in the browser and shows the accuracy

What happens to every message (respond() in both versions):

1. NLP pre-processing, entity extraction and intent classification.
2. If a task is pending (e.g. eligibility waiting for marks), the new facts fill the slot and the task continues.
3. Personal questions with applicant facts are sent to the forward-chaining engine (eligibility or scholarship rules).
4. The dialogue context resolves references ("it") and elliptical follow-ups ("What about MBA?").
5. The domain gate and the vector store handle unknown or low-confidence questions.
6. Otherwise the answer is filled from the frame KB. The response carries the text, intent, confidence, answer source and a full trace.

5. Dataset and Test-Case Preparation

There is no public dataset for a made-up university, so we wrote the knowledge and the test questions ourselves:

- **Knowledge sources:** 29 frames, 26 production rules, 18 intents with 109 example sentences, and 90 FAQ passages (75 generated automatically from the frames and 15 free-text passages).
- **Main test set:** 105 hand-written queries, deliberately different from the intent examples, in 8 categories that test different abilities. Each query is labelled with the expected intent and the expected answer key (e.g. program_fee:btech_ece, eligibility:mtech_ai:yes). An answer counts as correct only when the key matches exactly, so for reasoning questions the eligibility decision itself must be right.
- **Held-out test set:** 25 further queries written after development finished and evaluated only once, without any change to the system. This gives an unbiased estimate, because the main test set was also used to find bugs.

Category	Main set	Held-out	What it tests
factual	20	2	Direct factual questions
paraphrase	20	6	Same needs, different wording and synonyms
typo	10	3	Spelling mistakes ("hostle feez", "admision procedure")
reasoning	16	5	Personal eligibility / fee-after-scholarship decisions
multi_turn	14	3	5 conversations with follow-ups and slot filling
unstructured_faq	12	3	Free-text FAQ topics not modelled as frames
out_of_scope	10	3	Unrelated questions that must be refused
smalltalk	3	0	Greeting, thanks, goodbye
Total	105	25	

6. Performance Evaluation

Metrics used:

- **Answer accuracy:** share of queries whose answer key matches the expected key (main metric).
- **Intent accuracy and macro precision / recall / F1:** how well the question type is understood.
- **Out-of-scope precision / recall:** whether the chatbot refuses the right questions and only those.
- **Recall@k of the vector store:** whether the right free-text passage is among the top-k results.
- **Latency:** mean response time per query.

Metric	Hybrid KR + NLP

Answer accuracy	0.924
Intent accuracy	0.924
Intent macro precision	0.931
Intent macro recall	0.953
Intent macro F1	0.934
Out-of-scope precision	0.889
Out-of-scope recall	0.800
Mean latency (ms)	1.58
Vector store Recall@1 / Recall@3	0.92 / 1.00
Latency 95th percentile / max (ms)	2.23 / 3.31

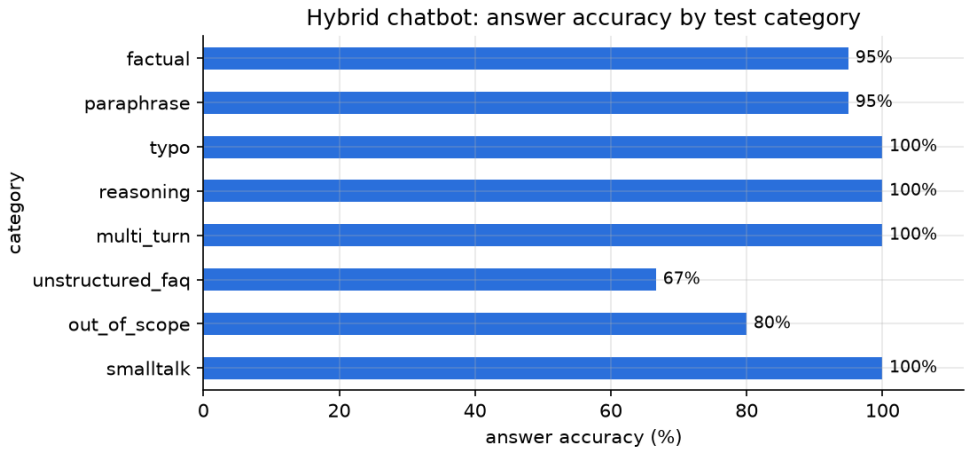


Figure 3. Answer accuracy of the hybrid chatbot by test category.

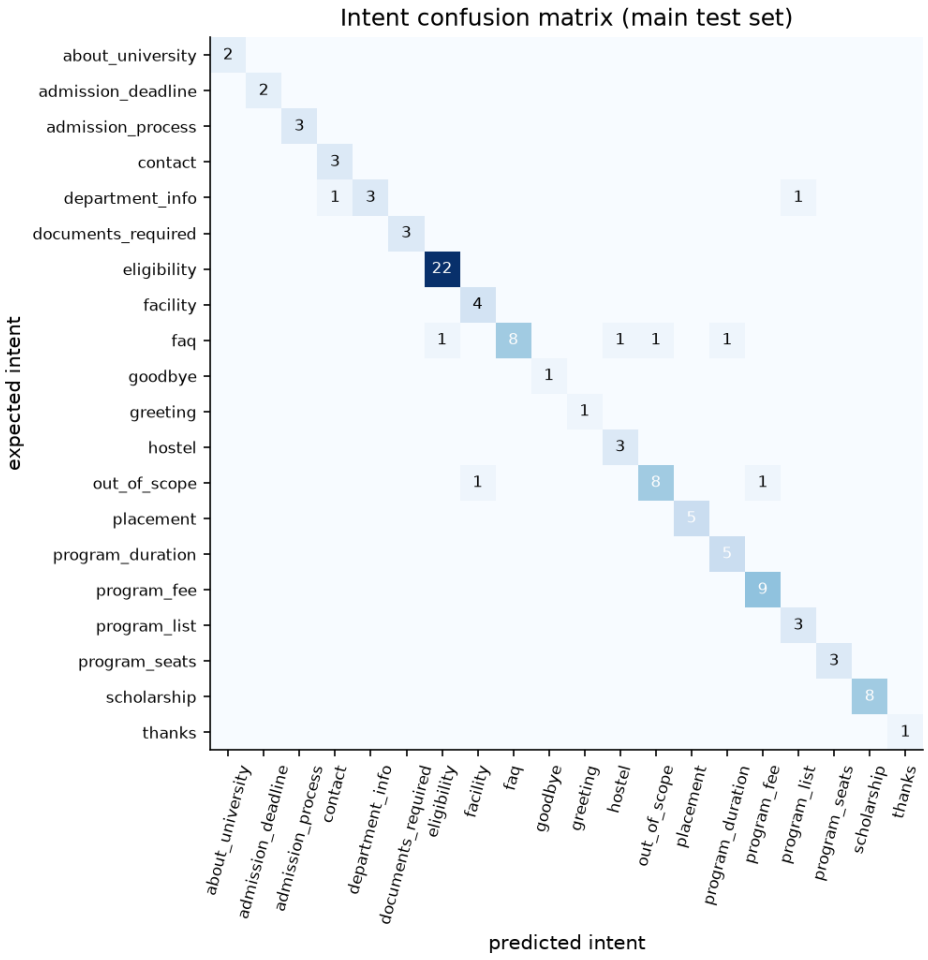


Figure 4. Intent confusion matrix on the main test set.

6.1 Error analysis

ID	Query	Predicted	Cause
F18	Which department offers MCA?	program_list:all	"offers" sent the query to the programme list instead of the department relation
P18	Where is the computer science department located?	contact	the word "located" triggered the contact intent
U02	Do you have wifi in the hostel?	hostel	"hostel" is a confident KB intent, so the Wi-Fi FAQ was never consulted
U03	Can diploma holders get lateral entry?	eligibility:ask_program	"diploma" was read as an applicant qualification, so the bot asked which programme
U08	Do we have to wear a uniform?	fallback	"uniform" has no lexical overlap with "dress code" (a limit of lexical embeddings)
U09	Can I study abroad for a semester?	program_duration:all	"semester" pulled the query towards the duration intent
O03	Can you book a movie ticket for me?	facility:all	"book" matched the library facility
O07	What is the share price of Infosys?	program_fee:all	"price" is a synonym of fee in the domain vocabulary

7. Comparison with Alternative Approaches

Two alternative chatbots were built on exactly the same knowledge: the same 90 FAQ question-answer pairs, with identical answer wording. Only the approach differs:

System	Description
Keyword FAQ matching (classic baseline)	Raw lower-cased text, word TF-IDF and cosine similarity to FAQ questions. No NLP, no KR, no reasoning, no memory.
NLP + vector retrieval	The same NLP normalisation and vector store as the proposed system, but every answer is retrieved from the FAQ corpus. No frames, rules or dialogue state.
Hybrid KR + NLP (proposed)	Frames + forward chaining + NLP + dialogue manager + vector-store fallback.

Metric	Keyword FAQ matching	NLP + vector retrieval	Hybrid KR + NLP (proposed)
Answer accuracy	0.314	0.381	**0.924**
Intent accuracy	0.419	0.552	**0.924**
Intent macro precision	0.482	0.578	**0.931**
Intent macro recall	0.432	0.502	**0.953**
Intent macro F1	0.388	0.490	**0.934**
Out-of-scope precision	0.077	0.167	**0.889**
Out-of-scope recall	0.100	0.400	**0.800**
Mean latency (ms)	0.32	0.71	**1.58**

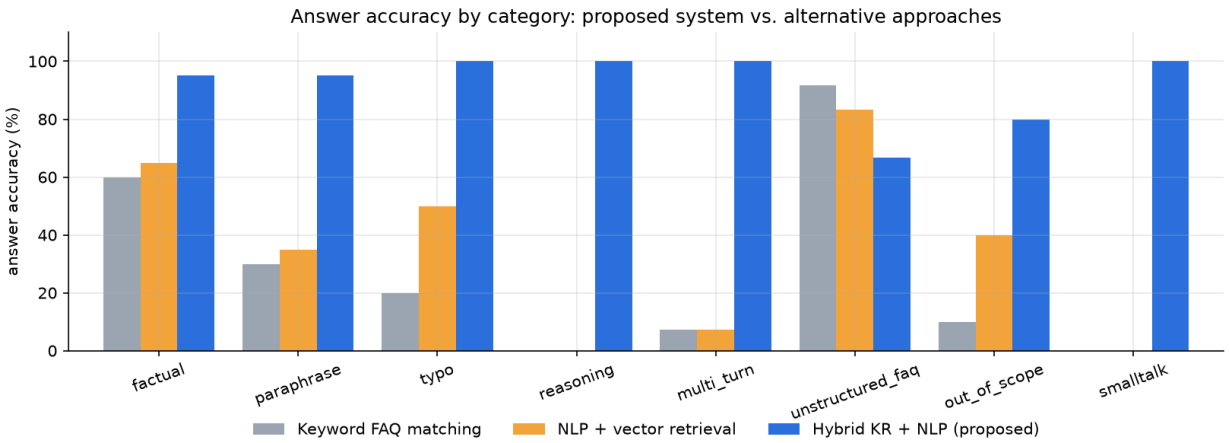


Figure 5. Answer accuracy by category: proposed system vs. alternative approaches.

7.1 Ablation study

To measure what each component contributes, one component was removed at a time:

Configuration	Overall	fact.	para.	typo	reas.	multi	FAQ	OOS
Full hybrid system	92.4%	95	95	100	100	100	67	80
without rule-based reasoning	72.4%	95	95	100	0	57	75	80
without dialogue context	83.8%	95	95	100	100	36	67	80
without spelling correction	86.7%	95	95	40	100	100	67	80
without vector-store fallback	84.8%	95	95	100	100	100	0	80

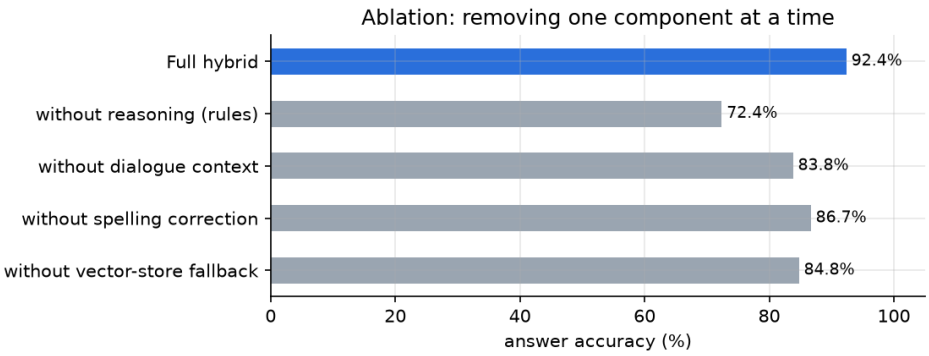


Figure 6. Ablation: answer accuracy when one component is removed.

7.2 Sensitivity to the confidence threshold

The intent-confidence threshold was fixed at 0.30 before testing. The sweep in Figure 7 shows that higher thresholds (0.35-0.55) give slightly higher accuracy and better out-of-scope recall on this test set. This was not adopted, because choosing a parameter on the test set would make the reported result optimistic. It should be validated on new data.

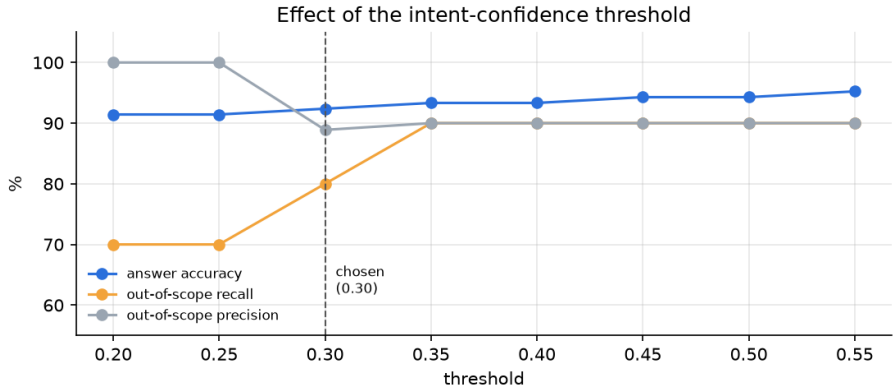


Figure 7. Effect of the intent-confidence threshold.

7.3 Held-out test set

Metric	Keyword FAQ matching	NLP + vector retrieval	Hybrid KR + NLP (proposed)
Answer accuracy	0.320	0.480	**0.800**
Intent accuracy	0.360	0.520	**0.800**
Intent macro F1	0.279	0.358	**0.670**
Out-of-scope recall	0.333	0.667	**0.333**

Note: while checking the held-out results, one label (H14, B.Sc Maths for M.Tech Data Science) was found to be wrong. The knowledge base accepts M.Sc Maths, not B.Sc, so the chatbot's "not eligible" answer was correct. The label was corrected; the system itself was not changed.

8. Results, Discussion and Conclusion

8.1 Results

- **Main test set:** on the 105-query main test set the hybrid chatbot answered 92.4% correctly (intent macro-F1 0.93), against 38.1% for NLP + vector retrieval and 31.4% for keyword FAQ matching.

- **Held-out set:** 80.0% for the hybrid chatbot, against 48.0% and 32.0%. The drop from the main test set is the expected optimism of a test set that was also used for debugging; the ranking of the three approaches is the same.
- **Speed:** mean 1.6 ms per answer on an ordinary laptop, without a GPU. The browser version feels just as instant.

8.2 Discussion and individual observations on the outputs

1. **The rules make the biggest difference.** Both alternatives scored 0% on reasoning questions and about 7% on multi-turn questions. They can show the general eligibility text, but they cannot say whether this particular student ("72% in B.E ECE") is eligible for M.Tech AI. When we switched off only the rule engine, accuracy fell from 92.4% to 72.4%, the biggest drop of any component.
2. **Inheritance kept the knowledge base small.** Duration, documents and deadlines are written once in the UG, PG and Research class frames, and every programme inherits them. When one of these values changes we edit one place. The trace also shows where a value came from, for example that mba.duration comes from the PGProgram frame.
3. **Explanations needed no extra work.** Each rule that fires writes one sentence, e.g. "[R3] A 5% relaxation applies for the OBC category, so the cut-off becomes 55%". On the website these sentences appear in the "How I answered" panel, so a student can see exactly why they are or are not eligible. A purely statistical chatbot cannot give this.
4. **Small NLP steps mattered more than we expected.** Spelling correction alone took typo accuracy from 40% to 100%. Without normalising B.E / B.Tech / M.Tech / plus two / CGPA, most entities would not have been found at all.
5. **People expect the bot to remember the conversation.** Without the context memory, multi-turn accuracy falls from 100% to 36%. Students ask "How long is it?" or reply to a question with just "I have 61 percent", and the bot has to understand that.
6. **Frames and free-text FAQs work well together, with one trade-off.** Without the vector store the free-text FAQ questions score 0%; with it, 67% (Recall@3 = 1.00). The plain retrieval chatbots still do better on these (83-92%), because our chatbot sometimes trusts a confident KB intent too much ("Wi-Fi in the hostel" goes to the hostel frame).
7. **Spotting out-of-scope questions is the weakest part.** The domain gate catches clearly unrelated questions (recall 80.0%), but questions that share words with the domain ("share price of Infosys", "book a movie ticket") still get a university answer.

8.3 Limitations

- Knowledge must be engineered by hand; rules cover only eligibility and scholarship policies.
- TF-IDF embeddings are lexical, so pure synonyms ("in charge of" vs "HOD", "uniform" vs "dress code") are missed.
- English only; the knowledge base is synthetic; the test sets were written by the developer, not collected from real users.

8.4 Conclusion

We designed, built and tested a university enquiry chatbot that combines knowledge representation with NLP. Frames with inheritance hold the university's facts. Forward-chaining rules answer personal eligibility and fee questions and explain each step. A simple NLP pipeline turns a free-form question into intent and entities, the dialogue manager handles follow-ups and missing details, and a vector store covers FAQ topics that are not in the frames. Our chatbot clearly beats the retrieval-only chatbots,

especially on reasoning and multi-turn questions. It stays fast and explainable, and as a website it can be demonstrated from any phone or laptop.

If we continue the project, we would like to: try a neural sentence embedding in place of TF-IDF (the embedder is already pluggable); learn the intent weights from real chat logs; add backward chaining for questions like "what do I need to become eligible?"; support Tamil and Hindi; and read the knowledge base from the real admissions database.

9. Demonstration of the Working Prototype

The working chatbot is online at https://manikandanayyakkannu.github.io/FAI-Mini-Project/. It opens straight away in any browser, on a laptop or a phone, with nothing to install.

9.1 The chat page

The page has three parts. On the left are sample questions grouped by topic, plus two guided demos that type out a short conversation on their own (useful during the presentation). The middle is the conversation. Under every answer a small label says which part of the system answered it. On the right, the "How I answered" panel shows the detected intent with its confidence and the top three intent scores, the rules that fired, the details picked out of the question, and the normalised text and key words. Clicking the label under any earlier answer brings its explanation back. After each answer the bot also suggests follow-up questions such as "How long is it?".

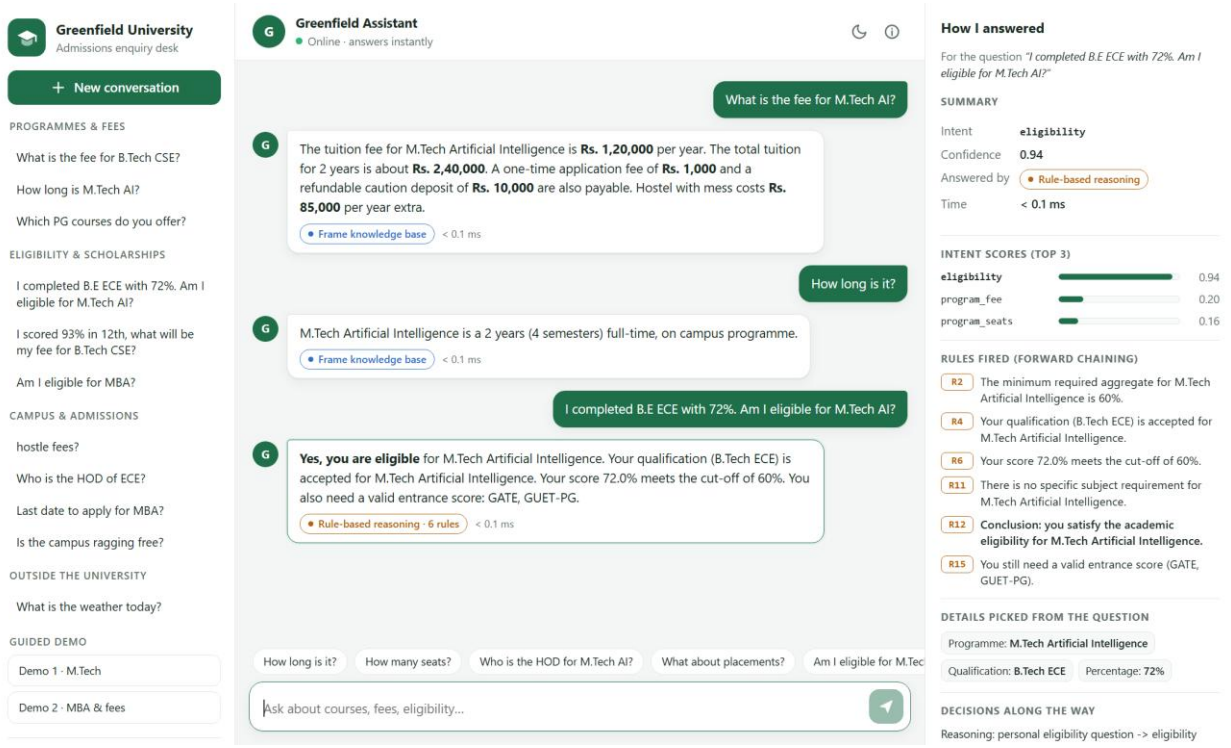


Figure 8. Demo 1: a follow-up question ("How long is it?") answered from context, then an eligibility decision with the rules that fired.

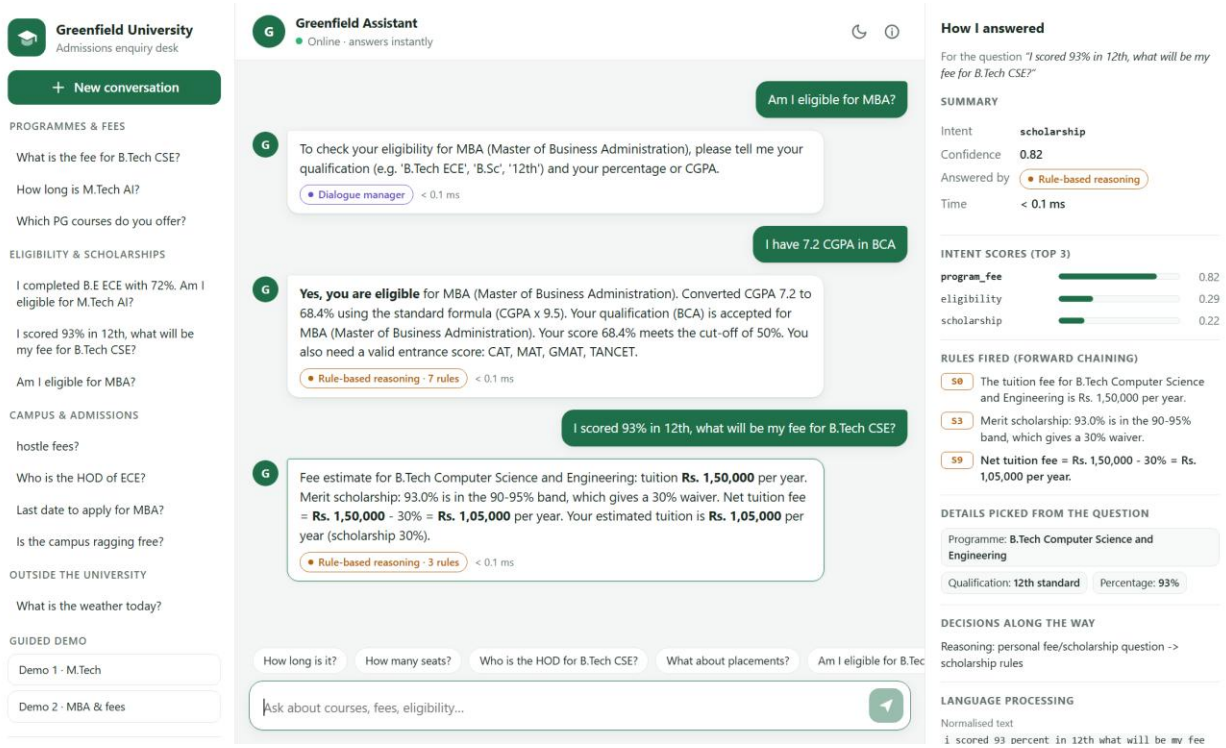


Figure 9. Demo 2: the bot asks for the missing details, converts CGPA to a percentage, and then works out a fee with the merit scholarship.

9.2 A sample conversation

This is one complete conversation with the chatbot. The line after each answer says which part of the system produced it.

```
You: What is the fee for M.Tech AI?
Bot: The tuition fee for M.Tech Artificial Intelligence is Rs. 1,20,000 per year. The total tuition for
2 years is about Rs. 2,40,000. A one-time application fee of Rs. 1,000 and a refundable caution deposit
of Rs. 10,000 are also payable. Hostel with mess costs Rs. 85,000 per year extra.
     (answered by: frame knowledge base)
You: How long is it?
Bot: M.Tech Artificial Intelligence is a 2 years (4 semesters) full-time, on campus programme.
     (answered by: frame knowledge base)
You: What about MBA?
Bot: MBA (Master of Business Administration) is a 2 years (4 semesters) full-time, on campus programme.
     (answered by: frame knowledge base)
You: Am I eligible for M.Tech AI?
Bot: To check your eligibility for M.Tech Artificial Intelligence, please tell me your qualification
(e.g. 'B.Tech ECE', 'B.Sc', '12th') and your percentage or CGPA.
     (answered by: dialogue manager)
You: I did B.E ECE with 6.8 CGPA and I qualified GATE
Bot: Yes, you are eligible for M.Tech Artificial Intelligence. Converted CGPA 6.8 to 64.6% using the
standard formula (CGPA x 9.5). Your qualification (B.Tech ECE) is accepted for M.Tech Artificial
Intelligence. Your score 64.6% meets the cut-off of 60%. Your GATE score is accepted. You will also get
a GATE stipend of Rs. 12,400 per month for GATE-qualified students.
     (answered by: rule-based reasoning)
You: I scored 96% in 12th and I am a state level athlete. What will my fee be for B.Tech CSE?
Bot: Fee estimate for B.Tech Computer Science and Engineering: tuition Rs. 1,50,000 per year. Merit
scholarship: 96.0% >= 95% gives a 50% tuition waiver. Sports quota (state/national level) adds a 25%
waiver. The total waiver is capped at 60%. Net tuition fee = Rs. 1,50,000 - 60% = Rs. 60,000 per year.
Your estimated tuition is Rs. 60,000 per year (scholarship 60%).
     (answered by: rule-based reasoning)
You: hostle fees for girls?
Bot: Separate hostels for boys (1,200 beds) and girls (900 beds) inside the campus. Rooms: single,
double and triple sharing. The fee is Rs. 85,000 per year, which includes accommodation and mess (veg
and non-veg). AC rooms cost Rs. 20,000 extra per year. In-time 9:00 PM for all students.
     (answered by: frame knowledge base)
```

```
You: Is the campus ragging free?
Bot: Greenfield University has zero tolerance for ragging. An anti-ragging committee and squad monitor
the campus and hostels 24x7, and students can report incidents anonymously on the helpline 1800-180-
5522.
     (answered by: vector-store search)
You: Who won the cricket match yesterday?
Bot: Sorry, I could not find an answer to that. I can help with programmes, fees, eligibility,
admissions, deadlines, documents, hostel, scholarships, placements and campus facilities. For anything
else please contact the admissions office at +91 44 2345 6789.
     (answered by: out of scope)
```

Appendix A. How to Use the Solution

1. **The chatbot:** open https://manikandanayyakkannu.github.io/FAI-Mini-Project/ in any browser. Click a sample question or type your own. "Demo 1" and "Demo 2" in the left panel play a short conversation for the presentation.
2. **Test results in the browser:** open tests/engine-test.html on the same site. It runs both test sets and shows the accuracy and every answer.
3. **The notebook:** open the project folder in VS Code (File > Open Folder), open University_Enquiry_Chatbot.ipynb, select the Python 3.12 kernel (top right) and click Run All. It recomputes every metric and figure in this report.
4. **Changing the knowledge:** the facts, rules, intents and FAQs live in data/*.json. Both the website and the notebook read these files, so an edit there changes both.

Appendix B. Software Used

All the software is free. Nothing needs to be installed to use the chatbot, since it runs in the browser. For the notebook we installed, for the Windows user only:

1. **Python 3.12.10** (winget install --id Python.Python.3.12 -e --scope user)
2. **Visual Studio Code** with the Python and Jupyter extensions
3. **Python libraries:** numpy, pandas, scikit-learn, matplotlib, jupyter and ipykernel (python -m pip install -r requirements.txt)
4. **Hosting:** GitHub Pages (Settings > Pages > Deploy from branch: main, folder / (root))

References

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